

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 1: Nutrition in Plants

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	1 — Nutrition in Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct.

Classic traps: treating respiration as switched off in daylight; reading a net gas measurement as if only one process were running; confusing which factor (chlorophyll vs light vs CO₂ vs water) a single controlled set-up actually isolates; assuming soil nitrogen is a raw material of the day's photosynthesis or that air nitrogen is directly usable; calling a parasite a saprotroph/symbiont; assuming a positive starch test proves the tissue is green; and mixing up xylem (water, up) with phloem (food, out from leaves). Do not write on this paper — mark answers on your answer sheet.

1. A sealed glass chamber holds one bean leaf in bright, steady light. Over a fixed hour the leaf takes in 18 units of carbon dioxide for one process while a second process inside the same leaf releases 6 units of carbon dioxide. A meter on the chamber wall measures only the NET change of the gas in the surrounding air. What does the meter read for carbon dioxide over that hour?

- (A) A net fall of 18 units, because respiration switches off in the light and only photosynthesis runs.
- (B) A net fall of 12 units, because photosynthesis used up more carbon dioxide than respiration released.
- (C) A net rise of 6 units, because only the carbon dioxide given out by the leaf reaches the air.
- (D) No change at all, because the two processes always cancel exactly inside a leaf.

2. In the same chamber as above, the experimenter now wants the carbon dioxide meter to read ZERO net change without removing the leaf or stopping either process. The single condition that achieves this is when:

- (A) the leaf is placed in total darkness so both processes halt and no gas moves.
- (B) the light is made as bright as possible, since maximum photosynthesis balances everything.
- (C) all the chlorophyll is removed so the leaf neither makes nor uses any gas.
- (D) the light is dimmed until photosynthesis takes in carbon dioxide at exactly the same rate respiration releases it.

3. A scientist tracks one carbon atom from the moment it enters a green leaf to the moment it becomes part of a starch grain stored in a potato. Which sequence is correct?

- (A) Carbon dioxide in air → absorbed by roots from soil → carried up in xylem → stored directly as starch.
- (B) Soil minerals → absorbed by roots → carried up in xylem → built into glucose → stored as starch.
- (C) Carbon dioxide in air → enters leaf through stomata → built into glucose in photosynthesis → carried as food in phloem → stored as starch.
- (D) Oxygen in air → enters through stomata → built into glucose in photosynthesis → stored as starch.

4. A pond's green algae gain 40 kilograms of new dry mass over a summer. A student insists 'almost all of that new dry weight must have come from minerals soaked out of the pond mud, since plants feed through their roots.' Choosing the best single correction based on this chapter:

- (A) Most of the new dry mass is water that the algae turned directly into solid plant body.
- (B) Most of the new dry mass came from oxygen the algae absorbed from the pond water.
- (C) The student is right: green plants build their bodies mainly from minerals taken in by their roots.
- (D) Most of the new dry mass is carbon and built from carbon dioxide captured during photosynthesis, not from soil minerals.

5. A student wants to prove that CARBON DIOXIDE specifically (not light, water or warmth) is the factor a leaf needs to form starch. Both leaves are on the same destarched plant, equally lit, watered and warmed for one sunny day. Which single difference between the two set-ups isolates carbon dioxide as the variable being tested?

- (A) One leaf's air has its carbon dioxide absorbed by potassium hydroxide while the other leaf, on the same plant, is left in ordinary air; everything else is identical.
- (B) One leaf is wrapped in foil while the other is left in the light.
- (C) One leaf is taken from a plant kept two days in the dark while the other is freshly destarched.
- (D) One leaf is variegated (part green, part white) while the other is fully green.

6. A teacher claims a single experiment can prove that BOTH light and chlorophyll are needed for photosynthesis. She takes ONE variegated leaf (green at the edges, white in the centre), covers half of its area — crossing both the green and white zones — with foil, exposes the destarched plant to a sunny day, then does the starch test. Which region turns blue-black, and does the claim hold?

- (A) The entire green border turns blue-black regardless of the foil, showing only chlorophyll matters.
- (B) The whole uncovered half turns blue-black including its white part, showing only light matters.
- (C) Only the part that is both green AND was left uncovered turns blue-black, so the one leaf does demonstrate both requirements.
- (D) No region turns blue-black, because covering any part of a variegated leaf stops all starch-making.

7. A copper beech tree has deep red leaves all summer yet grows tall and healthy in full sun, while a nearby variegated plant's WHITE leaf-patches never test positive for starch. A student asks why the red leaf clearly makes food but the white patch does not. The best explanation is:

- (A) Red pigment is a kind of chlorophyll, so any red or white tissue can photosynthesise equally.
- (B) The red leaf still contains chlorophyll hidden beneath a red pigment, so it photosynthesises; the white patch genuinely lacks chlorophyll, so it cannot.
- (C) The red tree gets its food from the soil, which is why white leaf-patches cannot.
- (D) White tissue makes food using white light while red tissue is purely decorative.

8. An aquatic plant in a beaker of water releases oxygen bubbles in light. A student counts 60 bubbles per minute with a lamp at 10 cm, 15 bubbles per minute at 20 cm, and almost none at 40 cm, keeping water, temperature and the plant the same. The single best conclusion is:

- (A) Moving the lamp away gradually destroys the plant's chlorophyll, so it makes fewer bubbles.
- (B) The rate of photosynthesis falls as the light reaching the plant decreases with distance, confirming light drives the process.
- (C) The plant runs out of water as the lamp moves away, so bubbling slows.
- (D) The bubbles are carbon dioxide, and less of it forms when the lamp is farther off.

9. A young branch on a tree is 'ringed': a complete strip of the outer phloem is peeled away all the way around, while the inner xylem is left intact. The leaves above the ring stay in full sun. Weeks later, what is the most likely observation and why?

- (A) The bark swells just above the ring as sugars pile up, because the leaves still get water through the xylem but the food they make cannot travel down past the broken phloem.
- (B) The leaves wilt and die within a day, because cutting the phloem stops water from reaching them.
- (C) The branch grows faster than ever, because removing phloem frees the leaves to photosynthesise more.
- (D) The leaves immediately stop making food, because the cut bark cuts off their carbon dioxide supply.

10. Surplus glucose in a leaf is rapidly converted to starch for storage. A student asks why the plant bothers to convert it rather than just keeping the glucose. The best reason, consistent with this chapter, is that starch:

- (A) is insoluble, so storing food as starch does not draw extra water into the cells the way dissolved sugar would.
- (B) can be made straight from sunlight without first making glucose.
- (C) is the gas the plant breathes out, so converting glucose helps it respire.
- (D) is the green pigment, so converting glucose to starch greens the leaf.

11. In the starch test a student boils the leaf in water, then dips it straight into iodine, skipping the alcohol step entirely. The leaf is genuinely full of starch, yet the blue-black colour is very hard to see against the leaf. The single best explanation is:

- (A) Skipping alcohol destroyed the starch, so no real colour formed.
- (B) Skipping alcohol left the green chlorophyll in the leaf, and its colour masks the blue-black starch reaction so it is hard to see.
- (C) Boiling in water alone removes all the starch, so nothing is left to detect.
- (D) Iodine cannot react at all unless alcohol is present to trigger it.

12. Before a photosynthesis experiment that ends in a starch test, the plant is kept in the dark for two days. A student proposes skipping this step to save time, arguing 'the iodine test will still detect whatever starch the experiment makes.' The flaw in skipping it is that:

- (A) without the dark period the leaf cannot absorb iodine during the test.
- (B) the dark period is what supplies the carbon dioxide the leaf will later use.
- (C) starch already stored in the leaf would also turn blue-black, so a positive result could no longer be credited to the experiment.
- (D) skipping it turns the leaf white and unable to photosynthesise.

13. A germinating pea seedling in a dark cupboard grows a pale, spindly shoot for several days, then yellows and dies. A second identical pea is moved into bright light on day 3 and instead grows sturdy and green. The pair together best show that:

- (A) light makes seeds germinate, which is why the dark seed eventually died.
- (B) seedlings photosynthesise faster in the dark, then switch off in the light.

(C) the dark seedling died only because the cupboard lacked water.

(D) the seed's own stored food powers early growth in the dark, but only light lets the seedling make new food and survive once that store is used up.

14. Cuscuta (a pale, leafless vine) and a mushroom on a dead log are both heterotrophs that lack chlorophyll, yet they are placed in different nutrition groups. A pale Cuscuta on a wire fence withers within days, while one on a living shrub thrives; the mushroom flourishes on dead wood and harms nothing living. The classification that fits both facts is:

(A) Cuscuta is a saprotroph living off dead matter such as the wire; the mushroom is a parasite on the log.

(B) Both are parasites, since both lack chlorophyll and take food from their surroundings.

(C) Cuscuta is a parasite that must take food from a living host; the mushroom is a saprotroph that feeds on dead, decaying matter.

(D) Cuscuta is a symbiont that helps its shrub; the mushroom is an autotroph making food in its cap.

15. A lichen is a fungus and an alga living together: the fungus supplies water, minerals and shelter, the alga supplies food. A student calls this 'a parasite, since the fungus lives off the alga.' The strongest reason the student is wrong is:

(A) The alga is actually harming the fungus, so the alga is the parasite instead.

(B) A lichen contains no living organisms, only minerals, so neither can be a parasite.

(C) Both partners benefit from the arrangement, so it is symbiosis, not parasitism in which one is harmed.

(D) The fungus makes the food and the alga merely stores it, so neither feeds off the other.

16. Four identical sealed transparent jars stand in bright light for a day, each holding one living thing kept moist: jar 1 a sprouting green pea seedling, jar 2 a germinating seed wrapped in foil so no light reaches it, jar 3 a fresh-cut but still-living green leaf, jar 4 an earthworm. In which ONE of the labelled jars is the carbon dioxide most likely to have RISEN by the end?

(A) Jar 2, the foil-wrapped germinating seed, because with no light it cannot photosynthesise and only respire, giving out carbon dioxide.

(B) Jar 1, the green pea seedling in light, because a lit green seedling gives out more carbon dioxide than it takes in.

(C) Jar 3, the cut but living green leaf in light, because it releases carbon dioxide and absorbs none.

(D) None of them, because every jar holds a living thing that uses up carbon dioxide.

17. A bean plant with Rhizobium root-nodules and a Venus flytrap growing in poor, boggy soil both end up with the nitrogen they need, but by different routes. Which statement correctly pairs each with its nitrogen source?

- (A) Both pull nitrogen gas straight from the air through their leaves.
- (B) The bean gets usable nitrogen from bacteria that fix air nitrogen in its root nodules; the flytrap gets nitrogen by digesting the insects it traps.
- (C) The bean digests insects in its nodules; the flytrap fixes nitrogen gas in its traps.
- (D) Both make their own nitrogen during photosynthesis from sunlight.

18. Two pots of the same plant are grown side by side outdoors in nitrogen-poor soil; both sit in the same nitrogen-rich air. Pot A is given a soil dressing of nitrogen-fixing bacteria; pot B is given none. Pot A greens up and grows well, while pot B stays pale and stunted. The single best conclusion is:

- (A) Air nitrogen became usable in pot A only because that plant opened larger stomata.
- (B) The bacteria in pot A removed the plant's need for nitrogen altogether.
- (C) The two pots prove nitrogen gas in air is a direct raw material of photosynthesis.
- (D) Plants cannot use nitrogen gas from the air directly; pot A thrived because the added bacteria fixed that nitrogen into a soil form the plant could absorb.

19. Two destarched plants are grown for several weeks: plant A in normal soil, plant B in soil sterilised free of nitrogen-fixing bacteria and given no fertiliser. Plant B grows pale and stunted. On one ordinary sunny day a still-greenish leaf from each is tested with iodine. What is observed, and what does it prove about nitrogen?

- (A) Only plant A turns blue-black, because nitrogen is one of the raw materials of photosynthesis.
- (B) Both leaves can turn blue-black; nitrogen is needed for healthy growth and green colour but is NOT a raw material in the day's photosynthesis.
- (C) Neither turns blue-black, because without nitrogen no plant can carry out photosynthesis at all.
- (D) Only plant B turns blue-black, because a nitrogen shortage forces the leaf to make extra starch.

20. A botanist insists a particular climbing plant is an autotroph. Four facts about it are gathered. Which ONE, on its own, would most strongly OVERTURN the autotroph claim?

- (A) Its tissues contain no chlorophyll at all, and threads from it pierce a host plant's stem to draw off ready-made food.
- (B) Its stems are bright green and it grows faster in stronger light.
- (C) It releases bubbles of oxygen when a piece of it is lit underwater.
- (D) It loses water vapour from its surface on warm afternoons.

21. A worksheet lists five living things and labels them all 'food-makers (autotrophs)': (i) a leafless Cuscuta vine, (ii) a green-stemmed cactus, (iii) a mushroom on a log, (iv) a floating green duckweed, (v) blue-green photosynthetic microbes. Exactly TWO labels are wrong. Which two, and why?

- (A) (ii) and (iv), because cactus stems and floating plants cannot photosynthesise.
- (B) (iii) and (v), because neither a mushroom nor a microbe can make food.

(C) (i) and (iii) — the leafless *Cuscuta* has no chlorophyll and steals food from a living host, and the mushroom has no chlorophyll and feeds on dead matter; neither makes its own food.

(D) (i) and (ii), because only large leafy trees are true autotrophs.

22. Trace the energy in a snake that ate a frog that ate a grasshopper that ate grass. One student claims the snake's energy 'comes mainly from the grass and ultimately from the sun.' Another claims it 'comes from the oxygen the snake breathes.' Who is right, and why?

(A) The second student, because oxygen is the fuel that powers animals.

(B) Both are equally right, since energy has two separate sources.

(C) Neither, because each animal makes its own energy from nothing.

(D) The first student: the grass captured sunlight as food and that energy passed up the chain, while oxygen only helps release energy already stored in food.

23. A student draws a food chain with arrows written as 'eagle → frog → insect → plant', explaining each arrow as 'feeds on'. A teacher says the arrows are fine for 'feeds on' but would point the OTHER way to show ENERGY FLOW. Which statement is correct?

(A) For energy flow the arrows must start at the plant: plant → insect → frog → eagle.

(B) Energy flow and 'feeds on' always point the same way, so nothing needs to change.

(C) Energy flows from the eagle downward, so the drawing already shows energy flow.

(D) Food chains carry no energy, so the arrow direction is meaningless.

24. A lake holds green phytoplankton, tiny shrimp that eat them, and herring that eat the shrimp. A pollutant kills almost all the phytoplankton over one week. Listing the events in the order they happen, which sequence is correct?

(A) Herring fall first → shrimp fall next → phytoplankton recover last.

(B) The phytoplankton (the producers) are gone → the shrimp soon starve and fall → the herring then run short and fall.

(C) The shrimp start photosynthesising to replace the lost phytoplankton, so nothing declines.

(D) The herring increase, because the water is now clearer for hunting.

25. A sealed glass globe holding a small green plant and moist soil has thrived in a sunny window for a month, its gases staying balanced. A child wraps it completely in black cloth for two weeks. Predicting the gas change inside and the plant's fate:

(A) Oxygen rises and the plant grows faster, because plants store oxygen in the dark.

(B) Nothing changes, because a sealed globe keeps its gases fixed regardless of light.

(C) Oxygen falls and carbon dioxide rises, because without light photosynthesis stops while respiration keeps using oxygen, so the plant weakens.

(D) The plant becomes a saprotroph and feeds on the soil, keeping the gases balanced.

26. A small mouse sealed alone in a jar soon becomes distressed as the air spoils, but a mouse sealed in a jar together with a green plant in BRIGHT LIGHT stays comfortable far

longer, because the plant adds oxygen. If the mouse-and-plant jar is instead kept in the DARK, then compared with the bright-light case the mouse will most likely become distressed:

- (A) sooner, because in the dark the plant makes no oxygen and, like the mouse, respire and uses up oxygen too.
- (B) later, because plants release extra oxygen in the dark.
- (C) never, because the plant alone keeps the air fresh whatever the light.
- (D) at exactly the same time, because plants have no effect on the air inside a jar.

27. A leaf disc from a plant kept 48 hours in the dark shows NO blue-black with iodine; a disc from the same plant taken after 6 hours of sunlight turns deep blue-black. What role does the dark-kept disc play in this experiment?

- (A) It proves that iodine simply fails to work on any leaf kept in the dark.
- (B) It is the negative control showing the destarched (starch-empty) baseline, against which the lit disc's newly made starch is judged.
- (C) It shows that two days of darkness permanently destroy the leaf's chlorophyll.
- (D) It is the source of the starch later found in the lit disc.

28. On a scorching, dry, windy afternoon a leaf's guard cells shut its stomata to limit water loss. Which of the following is the CORRECT chain of consequences for that leaf during the afternoon?

- (A) Stomata shut → sunlight is blocked from the chlorophyll → photosynthesis slows.
- (B) Stomata shut → chlorophyll is destroyed → photosynthesis stops for good.
- (C) Stomata shut → the roots stop taking up water → the leaf dies that afternoon.
- (D) Stomata shut → less water vapour escapes AND less carbon dioxide enters → photosynthesis slows for want of carbon dioxide.

29. A potted plant is weighed, then sealed at the soil so only its leaves are exposed, and left in a warm breeze; over hours its total weight steadily drops. When the same plant's leaf surfaces are all coated in petroleum jelly, the weight barely changes. The best explanation for the weight loss in the first case is:

- (A) the plant is burning its own mass into gas as it photosynthesises.
- (B) the soil is leaking water from the bottom of the pot.
- (C) water vapour is escaping from the leaves through the stomata (transpiration), and coating the leaf surfaces stops it.
- (D) the leaves are converting their stored starch directly into the lost weight.

30. A leaf on a well-watered plant in still, humid SHADE loses very little water vapour, yet it still makes starch whenever light reaches it. A nearby leaf on a hot, dry, windy day loses water vapour fast. Taken together these observations show that:

(A) transpiration is required for photosynthesis, so both should always rise and fall together.

(B) photosynthesis causes transpiration, so a leaf with low water loss cannot have made food.

(C) water loss and food-making are simply two names for the same single process.

(D) transpiration and photosynthesis are different processes, so reducing water loss does not stop food-making.

31. A clear plastic bag is tied over a living leaf on a sunny plant and water droplets collect inside; the same is done to a leaf whose cells have been killed, and far fewer droplets form. The best explanation is:

(A) the living leaf loses water vapour through its working stomata (transpiration); the dead leaf's stomata no longer function.

(B) the droplets are rainwater that leaked only into the living leaf's bag.

(C) the living leaf's xylem tubes squirt water directly out of the leaf surface.

(D) the dead leaf actively makes water inside the bag, which is why it has fewer droplets.

32. A student writes two claims: claim 1 — 'Xylem carries water and dissolved minerals upward from the roots'; claim 2 — 'Phloem also carries water upward, just more slowly.' How many of the two claims are correct, and which?

(A) Both claims are correct.

(B) Only claim 1 is correct; phloem carries food made in the leaves outward and largely downward, not water upward.

(C) Only claim 2 is correct; xylem actually carries the food.

(D) Neither claim is correct; both tissues carry food.

33. A pupil writes the photosynthesis word equation as 'carbon dioxide + water → (sunlight, chlorophyll) → glucose + carbon dioxide.' It contains exactly ONE mistake. The correction is:

(A) Change 'water' on the left to 'oxygen'.

(B) Remove 'chlorophyll' from above the arrow.

(C) Change 'glucose' to 'starch'.

(D) Change the right-hand 'carbon dioxide' to 'oxygen', since oxygen is the gas released.

34. Across the whole planet, animals breathing and fuels burning constantly remove oxygen from the air and add carbon dioxide to it, yet the proportions of these two gases in the atmosphere stay roughly steady over the years. Which process is mainly responsible for keeping that balance, and how?

(A) Photosynthesis in green plants, which takes in carbon dioxide and releases oxygen, replacing what breathing and burning use up.

(B) Respiration in green plants, which removes the extra carbon dioxide from the air.

(C) Transpiration, because the water vapour plants release washes carbon dioxide out of the air.

(D) The soil, which soaks up the extra carbon dioxide directly through plant roots.

35. Two destarched, well-watered green plants are given ample carbon dioxide and warmth for one day. Plant A sits under a bright lamp; plant B is sealed in a dark box. A leaf from each is then tested with iodine. The result, and what it isolates, is:

(A) Both turn blue-black, proving light is not needed.

(B) Only plant B (the dark box) turns blue-black, because darkness concentrates starch.

(C) Only plant A (under the lamp) turns blue-black, isolating LIGHT as the factor plant B lacked for making starch.

(D) Neither turns blue-black, because one day is too short for any plant to make starch.

36. In a greenhouse trial a crop is given identical water, warmth and carbon dioxide but four different light levels. Average new growth per week is: full sun 40 g, half-shade 22 g, deep shade 6 g, near-dark 0 g. The single best conclusion is:

(A) Light has no clear effect on growth, since every plot grew something.

(B) Darkness grows plants fastest, since the near-dark plot used the least energy.

(C) Growth rises with the amount of light, because more light drives more photosynthesis and so more food.

(D) The carbon dioxide, not the light, must have caused the differences.

37. A gardener notices her spinach is pale-yellow and stunted, though it gets full sun, ample water and air. Adding a nitrogen-rich manure turns the new leaves deep green and vigorous. This is best explained by the fact that nitrogen is needed to:

(A) act as the direct raw material the leaf turns into starch each sunny day.

(B) supply the carbon that photosynthesis builds into glucose.

(C) replace the sunlight the plant would otherwise lack.

(D) build proteins and the green pigment for healthy growth — though it is not itself a raw material of the day's photosynthesis.

38. In a legume–Rhizobium partnership each side both gives and gets. Which pairing correctly states what the BACTERIA receive from the plant?

(A) Sugary food and a protected home in the root nodules.

(B) Usable nitrogen compounds taken from the plant.

(C) Chlorophyll, so the bacteria can photosynthesise.

(D) Carbon dioxide, to build their own glucose.

39. A field cropped with maize year after year yields less as its nitrogen drains. Four remedies are proposed; which single one restores nitrogen NATURALLY, without manufactured fertiliser?

(A) Water the field more heavily each season.

(B) Plant the maize closer together to use the soil fully.

- (C) For one season grow a leguminous crop such as beans, whose root-nodule bacteria fix air nitrogen, then plough it back in.
- (D) Leave the field bare for a year so the nitrogen returns by itself.

40. In a lichen the fungus and the alga live together. If you wanted to point to the partner that actually **MAKES** the food the pair lives on, you would point to the:

- (A) fungus, because its threads photosynthesise even in the dark.
- (B) alga, because it has chlorophyll and photosynthesises, making the food for both partners.
- (C) fungus, because it absorbs ready-made food from the air for both.
- (D) neither, because a lichen lives only on minerals and needs no food-maker.

41. A pupil tests four foods with iodine: boiled rice, a slice of apple, raw potato, and a piece of boiled egg yolk. The rice and potato turn deep blue-black; the apple and egg yolk barely change. The safest conclusion is:

- (A) Only living, uncooked foods can turn blue-black with iodine.
- (B) The blue-black colour shows the rice and potato contain chlorophyll.
- (C) Iodine turns blue-black only with foods that taste sweet.
- (D) Rice and potato are rich in starch, while the apple and egg yolk contain little or none.

42. A leaf's broad, flat shape (rather than thick and rod-like) and the crowding of its chlorophyll-rich cells just beneath the sunlit upper skin together serve one design principle. That principle is that the leaf is built to:

- (A) present the largest possible surface to sunlight and place the chlorophyll where the light is strongest, for efficient photosynthesis.
- (B) store the maximum amount of water for dry spells.
- (C) be too tough for any insect to chew. (D) stay heavy so the wind cannot move it.

43. Seen under a microscope, a leaf's lower surface is dotted with tiny pores, each able to open and close between a pair of bean-shaped guard cells, while its upper surface has very few. These pores are the leaf's main gateway for exchanging gases with the air. What are they called, and what is their chief role in nutrition?

- (A) Veins; they carry the food made in the leaf out to the rest of the plant.
- (B) Root hairs; they take in the water that photosynthesis needs.
- (C) Stomata; they let carbon dioxide enter for photosynthesis (and let oxygen and water vapour pass out).
- (D) Stomata; they absorb mineral nutrients straight from the air for the plant.

44. Carrots store food in their roots, potatoes in underground stems and wheat grains in their seeds — yet in every case the food was first made in the green leaves. Which statement correctly describes how that food reaches these very different storage parts?

- (A) Each storage part makes its own food directly, since food can be made anywhere in a plant.
- (B) Food made in the leaves travels through the phloem to the storage parts, where the surplus is stored as starch.
- (C) The roots absorb ready-made food from the soil and pass it up to the other parts.
- (D) Water drawn up through the xylem is turned into stored food inside each part.

45. Which single statement about the starch test is FALSE?

- (A) A blue-black colour with iodine indicates that starch is present.
- (B) A brown (no-colour) result always means the tissue never contained any chlorophyll.
- (C) Destarching the plant beforehand lets any new starch be credited to the experiment.
- (D) Boiling the leaf in alcohol removes chlorophyll so the colour change is easy to read.

46. In the starch test the leaf is boiled in alcohol by standing the alcohol tube in a beaker of hot water, rather than heating the alcohol directly over a flame. The main reason is that:

- (A) water added to the alcohol is what dissolves the chlorophyll.
- (B) direct heating would turn the alcohol into starch.
- (C) it is the flame's light, not its heat, that the alcohol step needs.
- (D) alcohol is highly flammable, so heating it directly over a flame is dangerous; standing the tube in hot water heats it safely.

47. Both a banyan tree and a goat need food, but they obtain it in fundamentally different ways. The essential difference is that the banyan:

- (A) eats small soil creatures, whereas the goat eats leaves.
- (B) makes food from sunlight inside its roots, whereas the goat makes food in its stomach.
- (C) builds its own food from carbon dioxide and water using sunlight (autotroph), whereas the goat must consume ready-made food (heterotroph).
- (D) and the goat both take ready-made food from their surroundings.

48. Left damp and warm for a few days, a slice of bread grows a furry mould — a fungus that has no chlorophyll of its own. How does this fungus obtain its food from the bread?

- (A) It photosynthesises using the light in the kitchen.
- (B) It secretes digestive juices onto the bread that break the food down, then absorbs the dissolved food — feeding as a saprotroph.
- (C) It traps and digests tiny insects that land on the bread.
- (D) It makes its own food from the warmth and moisture around it.

49. A leafless, pale-yellow *Cuscuta* is transplanted onto a plastic garden stake; within a week it shrivels and dies, but a cutting placed on a living hibiscus bush spreads and thrives. The behaviour proves *Cuscuta* is:

- (A) a saprotroph that needs dead matter such as the plastic to feed on.
- (B) a green autotroph that has only briefly lost its colour.
- (C) a symbiont that helps whatever object it climbs upon.
- (D) a parasite that can survive only by drawing food from a living host.

50. A sundew is a small green plant of nitrogen-poor bogs whose sticky leaves trap and digest insects. Which single statement correctly accounts for BOTH its green leaves and its insect-trapping?

- (A) The green leaves are merely ornamental; all of its food comes from the insects.
- (B) It photosynthesises with its green leaves to make energy-rich food, and traps insects mainly to obtain the scarce nitrogen its boggy soil lacks.
- (C) It is a parasite, living on the insects it traps.
- (D) It is a saprotroph that feeds only on dead, decaying matter in the bog.

51. A green plant is sealed in a transparent box and kept in steady bright light for several hours, with photosynthesis clearly outpacing respiration. Comparing the air at the end with the air at the start, which change is expected?

- (A) Oxygen has fallen and carbon dioxide has risen.
- (B) Both oxygen and carbon dioxide have risen.
- (C) Nitrogen has fallen sharply while oxygen is unchanged.
- (D) Oxygen has risen and carbon dioxide has fallen.

52. Food chains are drawn starting with a green plant, never with a mushroom, even though both are alive. The reason a green plant — and not the mushroom — belongs at the start is that the green plant:

- (A) is the only living thing in the chain that respire.
- (B) is eaten by the mushroom to begin the chain.
- (C) captures sunlight to create brand-new food, introducing fresh energy, while the mushroom only breaks down food that already existed.
- (D) is larger than the mushroom and so comes first.

53. Tigers eat only meat, never plants, yet biologists say even a tiger ultimately depends on green plants. The strongest reason is that:

- (A) tigers occasionally nibble grass when they are ill.
- (B) green plants give the tiger its minerals directly from the soil.
- (C) plants release most of their oxygen at night, when the tiger hunts.
- (D) the deer a tiger eats fed on plants, so the energy in the tiger's food was first captured from sunlight by plants, which also supply the oxygen it breathes.

54. A pond's green algae double in mass over summer while a snail population in the same pond also grows by eating those algae. If, instead, a cloud of fine silt permanently shades the pond so almost no light reaches the algae, the chain of consequences over the following weeks is best described as:

- (A) the snails thrive because shaded algae become softer and easier to eat.
- (B) the algae switch to feeding on the silt and grow even faster in the shade.
- (C) the algae photosynthesise little, so they decline; the snails that feed on them then run short of food and also decline.
- (D) nothing changes, because algae do not actually need light to make food.

55. A pupil argues: 'Autotrophs make their own food, so a green plant locked in a sealed lightless cupboard would feed itself indefinitely.' The single best correction is:

- (A) Wrong — even autotrophs need raw materials and energy from outside (carbon dioxide, water and especially light); in the dark it cannot make food and will starve.
- (B) Correct — autotrophs need nothing from outside once they are grown.
- (C) Wrong — the plant would survive by eating the cupboard's wood.
- (D) Wrong — the plant would switch to taking ready-made food from the air.

56. A destarched plant has one leaf sealed in CO₂-free air (using potassium hydroxide) and another leaf left in normal air; both are on the same plant in bright light for six hours, then tested with iodine. A student writes that 'the leaf in CO₂-free air turned blue-black because removing a gas speeds up food-making.' What is the correct result, and what error did the student make?

- (A) Both leaves turn blue-black, because light alone is enough to make starch.
- (B) Only the normal-air leaf turns blue-black; the student wrongly treated carbon dioxide as a hindrance when it is actually a needed raw material.
- (C) Neither leaf turns blue-black, because destarching destroyed their ability to ever make starch.
- (D) Only the CO₂-free leaf turns blue-black, exactly as the student predicted, because KOH feeds the leaf.

57. A scientist records a single leaf's net oxygen exchange over 24 hours: net oxygen is released from about 7 a.m. to 6 p.m. and net oxygen is taken IN the rest of the time, crossing zero at exactly two instants around dawn and dusk. What is happening at each of those two zero-crossing instants?

- (A) The leaf has briefly stopped both photosynthesis and respiration.
- (B) The leaf is only respiring and is using up all of its own oxygen internally.
- (C) Photosynthesis has completely stopped while respiration carries on at full speed.
- (D) The rate of oxygen made by photosynthesis exactly equals the rate of oxygen used by respiration.

58. Fungi and bacteria that feed on fallen leaves, dead wood and animal remains are often called 'nature's decomposers,' even though they make no food of their own. Why are these saprotrophs important for nutrition in the living world?

- (A) They break down dead matter and return its locked-up nutrients to the soil, where plants can absorb and reuse them.
- (B) They make food from sunlight and pass it on to the plants around them.

- (C) They are the source of the oxygen that green plants need for photosynthesis.
- (D) They have no real importance, since only the food-makers in a habitat matter.

59. A girl reasons: 'When a tree grows from a tiny seed into a heavy trunk, all that extra solid wood must have been lifted up by the roots from the soil as solid material.' Using what this chapter teaches about how plants make food, the best correction is:

- (A) Most of the added solid material is built from carbon dioxide taken from the air during photosynthesis, not lifted as solid matter from the soil.
- (B) Most of the added wood is solid soil drawn up whole through the roots and hardened.
- (C) Most of the added wood comes from oxygen the tree absorbed and turned into solid matter.
- (D) The girl is right: a tree's whole body is solid soil carried up by the roots over the years.

60. Which single row correctly matches each transport job to its tissue and direction?

- (A) Water: phloem, upward — and — food: xylem, downward.
- (B) Water and minerals: xylem, upward from the roots — and — food (sugars): phloem, away from the leaves and largely downward.
- (C) Both water and food travel in the xylem only, in both directions.
- (D) Water: xylem, downward to the roots — and — food: phloem, upward to the leaves.

Solutions — Science (Class 7), Chapter 1:

Nutrition in Plants — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	B	21	C	31	A	41	D	51	D
2	D	12	C	22	D	32	B	42	A	52	C
3	C	13	D	23	A	33	D	43	C	53	D
4	D	14	C	24	B	34	A	44	B	54	C
5	A	15	C	25	C	35	C	45	B	55	A
6	C	16	A	26	A	36	C	46	D	56	B
7	B	17	B	27	B	37	D	47	C	57	D
8	B	18	D	28	D	38	A	48	B	58	A
9	A	19	B	29	C	39	C	49	D	59	A
10	A	20	A	30	D	40	B	50	B	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Both processes run at once in the lit leaf. Photosynthesis pulls 18 units of CO₂ out of the air; respiration puts 6 units back. The meter sees the NET effect on the air: 18 removed minus 6 added = 12 removed, i.e. a net fall of 12 units. The claim that respiration switches off in light is the day-respiration misconception — respiration never stops, so 18 is wrong. The 'net rise of 6' answer counts only the respiratory output and forgets the much larger photosynthetic uptake from the same air. 'No change/always cancel' is only true at the exact compensation point where the two rates happen to be equal; here uptake clearly exceeds release, so the gases do not cancel.

2. (D) A zero NET reading means CO₂ taken in by photosynthesis equals CO₂ given out by respiration — the compensation point. Lowering light reduces photosynthesis until its uptake rate drops to match the steady respiratory output, giving zero net change while both processes still run. Total darkness does NOT halt both processes: respiration continues, so the meter would show a net RISE, not zero. Maximum brightness drives photosynthesis far above respiration, giving a large net FALL, not balance. Removing

chlorophyll stops photosynthesis entirely while respiration continues, again giving a net rise — and it does stop a process, which the question forbids.

3. (C) The carbon of plant food comes from carbon dioxide, which enters the LEAF through stomata (not the roots/soil), is fixed into glucose during photosynthesis, then travels as food through the PHLOEM to storage organs where surplus glucose is stored as starch. The first wrong option mislocates CO₂ uptake at the roots and uses xylem (which carries water, not food). The minerals option confuses soil minerals with the carbon source — minerals are needed for health but are not the carbon backbone of glucose. The oxygen option names the wrong gas: oxygen is a PRODUCT released, not the raw material that supplies carbon.

4. (D) The bulk of a plant's dry body is made of glucose/starch and related substances whose carbon comes from carbon dioxide fixed in photosynthesis — so the new dry mass is mostly carbon captured from CO₂, not soil minerals. Water is taken in, but it is the source of hydrogen, not the bulk solid dry mass attributed here, and 'turning water directly into solid body' misstates the process. Oxygen is a product RELEASED by photosynthesis, not a building block absorbed to add mass. The student's root-feeding idea is the classic soil-as-food misconception the chapter explicitly overturns: roots supply water and minerals, but food (and thus most dry mass) is made in the leaves from CO₂.

5. (A) A controlled experiment isolates one factor by changing ONLY that factor between two otherwise-identical set-ups. To prove carbon dioxide specifically is needed, keep light, water, warmth and the plant identical and differ ONLY in CO₂: one leaf's air has its CO₂ absorbed by potassium hydroxide while the other, on the same plant, stays in ordinary air. Any difference in the starch test can then be pinned on CO₂ alone. Wrapping one leaf in foil changes LIGHT, not CO₂. A plant kept two days in the dark differs in light/destarching history, not CO₂. A variegated leaf (green vs white) tests CHLOROPHYLL, not carbon dioxide.

6. (C) Starch forms only where ALL requirements are met. Photosynthesis needs chlorophyll (so white tissue fails) and light (so foil-covered tissue fails). The only region satisfying both is green tissue that was also lit — it alone turns blue-black. Comparing the lit-green region with the lit-white region shows chlorophyll is needed; comparing the lit-green region with the covered-green region shows light is needed — so one leaf demonstrates both. 'Entire green border regardless of foil' ignores the light requirement on the shaded green. 'Whole uncovered half including white' ignores the chlorophyll requirement on the white tissue. 'No region at all' is wrong because the lit-green tissue still has everything it needs.

7. (B) The decisive factor is the presence of chlorophyll, the light-capturing pigment. A red leaf only LOOKS non-green because a red pigment masks the chlorophyll that is still present, so it photosynthesises and grows. A white patch contains no chlorophyll at all, so it cannot make food and tests negative for starch — exactly the difference observed. Calling

red pigment 'a kind of chlorophyll' is false; colour alone does not equal chlorophyll. The soil-feeding answer revives the food-from-soil misconception. 'White tissue makes food using white light' is backwards — colourless tissue without chlorophyll cannot photosynthesise no matter the light.

8. (B) Light reaching the plant weakens sharply with distance, and the bubble rate (a measure of photosynthesis) drops in step — strong evidence that light intensity controls the rate of photosynthesis. Only light was varied; water, temperature and the plant were held constant, so those cannot be the cause. Chlorophyll is not destroyed by simply dimming the light; the chlorophyll is intact and works again when the lamp returns. The plant is sitting in water throughout, so it cannot 'run out of water.' The bubbles are oxygen (a product released in light), not carbon dioxide — CO₂ is taken in, not bubbled out, during photosynthesis.

9. (A) Xylem (intact) carries water UP to the leaves, so the leaves keep their water and keep photosynthesising. Phloem carries the FOOD made in the leaves DOWN/out; with the ring broken, that food cannot pass the cut and accumulates as a swelling just above the ring. The wilt-and-die option confuses the two tissues — water travels in the untouched xylem, so water supply is fine. 'Grows faster' is baseless; removing phloem hinders, not helps, the plant. The carbon-dioxide option is irrelevant: CO₂ enters through stomata on the leaves, which the ringing does not touch.

10. (A) Glucose is soluble; large amounts dissolved in cells would upset the cell's water balance by drawing in water. Converting it to insoluble starch lets the plant stockpile food compactly without that osmotic disturbance — which is why surplus glucose is stored as starch. Starch cannot be made directly from sunlight; glucose is made first in photosynthesis, then converted. Starch is a solid food store, not a gas the plant breathes out — that confuses it with carbon dioxide. Starch is not a pigment; the green pigment is chlorophyll, a different substance entirely.

11. (B) Boiling the leaf in alcohol is the step that removes the green chlorophyll; skipping it leaves the chlorophyll in place. Iodine still turns the starch blue-black, but that dark colour is hidden behind the strong green of the leaf, so it is very hard to see — the starch is really there, just masked. Skipping alcohol does NOT destroy the starch; alcohol removes pigment, it does not break down food. Boiling in water alone does not wash out the starch, so 'nothing left to detect' is wrong. Iodine reacts with starch directly and needs no alcohol to 'trigger' it; alcohol's only job is to strip the masking chlorophyll so the colour can be seen.

12. (C) The plant is kept in the dark first so its leaves use up their stored starch (destarching). Then any starch found at the end must be NEW starch made during the experiment, so a positive iodine result can be credited to the conditions being tested. Skip the dark period and the leaf may still hold old, stored starch, which would also turn blue-black — a false positive that ruins the conclusion. The dark period has nothing to do with

iodine absorption, and it does not 'supply carbon dioxide' — CO₂ comes from the air during the lit phase. It does not turn the leaf white either; darkness empties starch, it does not destroy chlorophyll or the ability to photosynthesise.

13. (D) A seed carries its own food store, which fuels early growth even in the dark — that is why the cupboard seedling grows at first. But it cannot make NEW food without light (photosynthesis needs light), so once that store runs out it yellows and dies. The second pea, moved into light on day 3, starts photosynthesising and survives. So the pair shows: stored food powers early dark growth, but only light lets the seedling make food and survive once the store is gone. Light does not 'make seeds germinate' here — both germinated; the difference is survival afterwards. Seedlings do not photosynthesise faster in the dark — they cannot photosynthesise at all without light. The dark seedling did not die merely from lack of water; the lit one had the same water and lived.

14. (C) Cuscuta dies on a non-living wire but thrives on a living shrub — proof it needs a LIVING host, the mark of a parasite. The mushroom thrives on DEAD wood and harms nothing alive — the mark of a saprotroph that digests dead, decaying matter. Calling Cuscuta a saprotroph contradicts its death on the lifeless wire (it cannot feed on dead matter). Lumping both as parasites ignores that the mushroom needs no living host. The symbiont/autotroph option is doubly wrong: Cuscuta harms its host (not mutual benefit), and the mushroom has no chlorophyll, so it cannot be an autotroph making food in its cap.

15. (C) In a parasitic relationship one partner is harmed; in the lichen, the fungus gains food while the alga gains water, minerals and shelter — a mutually beneficial partnership, i.e. symbiosis, not parasitism. The 'alga harms the fungus' option invents harm where the description states mutual benefit. 'No living organisms, only minerals' is false — both the fungus and the alga are living. The role-swap option is factually wrong: the ALGA (with chlorophyll) makes the food, not the fungus, so it misstates who feeds whom and still misses the key point that both benefit.

16. (A) Carbon dioxide RISES wherever an organism only respire (gives out CO₂) and does not photosynthesise to take it back. Jar 2 holds a germinating seed wrapped in foil — no light, so no photosynthesis; it only respire, so its CO₂ rises. Jar 1 (a lit green pea seedling) and Jar 3 (a lit, cut-but-living green leaf) both photosynthesise in the light, taking in more CO₂ than they release, so their CO₂ FALLS, not rises. 'None of them' is wrong because the foil-wrapped seed clearly raises CO₂. Among the labelled options offered, only Jar 2 fits.

17. (B) Each plant solves a nitrogen shortage by a different route. The bean cannot use nitrogen gas directly; Rhizobium bacteria in its root nodules fix air nitrogen into a usable compound for it. The Venus flytrap, in nitrogen-poor bog, digests the insects it traps to obtain nitrogen. Neither pulls nitrogen gas straight from the air through its leaves — plants cannot use N₂ directly. The roles are not swapped: the flytrap traps insects, the bean uses

nodule bacteria. And nitrogen is not 'made during photosynthesis' — photosynthesis builds glucose from CO₂ and water, not nitrogen.

18. (D) Both pots sit in the same nitrogen-rich AIR, yet only pot A (given nitrogen-fixing bacteria) thrives — proof that plants cannot use atmospheric nitrogen gas directly. The bacteria fix that nitrogen into a soil form the plant's roots can absorb, so pot A greens up while pot B stays starved. Stomatal size has nothing to do with nitrogen uptake. The bacteria did not 'remove the plant's need for nitrogen' — they supplied it in usable form. And the two pots do NOT show nitrogen gas is a raw material of photosynthesis; photosynthesis uses CO₂, water and light — nitrogen is for growth, not the day's food-making.

19. (B) Photosynthesis on a given day needs carbon dioxide, water, light and chlorophyll — NOT soil nitrogen. So as long as plant B's leaf still has some chlorophyll, it can make starch that day and turn blue-black, just like plant A. The long-term paleness and stunting show nitrogen matters for healthy growth and green colour, but the same-day starch test shows it is not a direct raw material of photosynthesis. 'Only A turns blue-black' wrongly treats nitrogen as a photosynthetic raw material. 'Neither' overstates nitrogen's role — a green-ish leaf still photosynthesises. 'Only B, forced to make extra starch' is backwards: nitrogen shortage hampers growth, it does not boost starch-making.

20. (A) An autotroph makes its own food using chlorophyll and light. The single fact that an organism has NO chlorophyll and sends threads into a host to draw off ready-made food shows it cannot make its own food — it is a parasite, which directly overturns the autotroph claim. Bright green stems that grow faster in stronger light actually SUPPORT being an autotroph. Releasing oxygen bubbles when lit underwater is a hallmark of photosynthesis, again supporting autotrophy. Losing water vapour on warm afternoons is ordinary transpiration and says nothing about how the plant feeds.

21. (C) An autotroph makes its own food, which needs chlorophyll. (i) A leafless *Cuscuta* has no chlorophyll and steals food from a living host — a parasite, so its autotroph label is wrong. (iii) A mushroom has no chlorophyll and feeds on dead matter — a saprotroph, so its label is wrong too. The green-stemmed cactus (ii), green duckweed (iv) and photosynthetic blue-green microbes (v) all contain chlorophyll and make their own food, so they ARE autotrophs. The two wrong labels are therefore (i) and (iii). Options blaming the cactus, duckweed, microbes, or claiming 'only large leafy trees' are autotrophs misjudge which organisms truly photosynthesise.

22. (D) A food chain's energy traces back to the green plant, which captured sunlight and stored it as food; that energy passes up the chain as grass → grasshopper → frog → snake. So the first student is right: the snake's energy comes ultimately from the sun via the grass. Oxygen (the second student's answer) is only used to RELEASE energy already stored in food during respiration — it is not the source of that energy. Energy does not have two

separate sources, and no animal 'makes energy from nothing' — that would break the conservation the chapter teaches.

23. (A) The 'feeds on' arrows the student drew (eagle → frog → insect → plant) point from eater to eaten. ENERGY flow points the OTHER way — from the producer that captures sunlight up to the top consumer: plant → insect → frog → eagle. So the corrected energy-flow chain starts at the plant. Energy flow and 'feeds on' do NOT point the same way. Energy does not flow 'from the eagle downward' — the eagle is the end of the line, not the source. Food chains certainly do carry energy, so the arrow direction is not meaningless.

24. (B) The phytoplankton are the producers at the base of the chain. Kill them and the shrimp that eat them lose their food first and decline; the herring that eat the shrimp then run short and decline next. So the collapse passes UP the chain from the base. The shrimp cannot start photosynthesising to replace the lost producers — they are animals. The herring do not increase from 'clearer water'; losing the base of the chain harms everything above it. And the order is not herring-first — the impact begins at the producers and rises.

25. (C) In light the sealed globe balanced because photosynthesis used the CO₂ from respiration and respiration used the O₂ from photosynthesis. Wrapped in black cloth, photosynthesis stops but respiration continues, so the plant keeps consuming oxygen and releasing carbon dioxide with nothing to recycle them: O₂ falls, CO₂ rises, and the plant weakens. Oxygen cannot rise in the dark — there is no photosynthesis to make it. 'Nothing changes' wrongly assumes a sealed globe freezes its gases; respiration keeps altering them. A green plant does not become a saprotroph feeding on the soil.

26. (A) In bright light the plant photosynthesises and adds oxygen, helping the mouse. In the dark the plant makes NO oxygen and, like the mouse, respire and uses oxygen up — so now two organisms compete for the same limited air, and the mouse becomes distressed SOONER than in the light. Plants do not release 'extra oxygen in the dark' — photosynthesis needs light. The plant does not keep the air fresh regardless of light. And the plant certainly does affect the jar's air: in the dark it actively consumes oxygen.

27. (B) The dark-kept disc is the NEGATIVE CONTROL. Forty-eight hours of darkness lets the leaf use up its stored starch, so it tests negative — establishing a starch-empty baseline. The disc taken after 6 hours of sunlight has made fresh starch and tests positive, so its blue-black is unambiguously NEW starch. Iodine does not 'fail on dark leaves' — the dark disc was negative because it had no starch, not because iodine stopped working. Two days of dark do not permanently destroy chlorophyll — the lit disc proves the leaf still photosynthesises. And the dark disc is not the source of the lit disc's starch.

28. (D) Stomata are the pores through which both water vapour leaves and carbon dioxide enters. Shutting them to save water therefore ALSO cuts off the CO₂ supply, and since CO₂ is a raw material of photosynthesis, food-making slows for want of it. Sunlight is

not blocked by closing the tiny pores — light passes through the leaf surface regardless. Closing stomata does not destroy chlorophyll. It reduces water LOSS, but it does not make the roots stop taking up water or kill the leaf that afternoon — the real consequence is the CO₂ cut-off slowing photosynthesis.

29. (C) The plant is sealed at the soil, so the only route for the steady weight loss is water vapour escaping from the leaves through the stomata — transpiration. Coating every leaf surface with petroleum jelly blocks the stomata, so the loss almost stops, confirming the leaves were the exit. The plant is not 'burning its own mass' in photosynthesis — photosynthesis adds mass. The soil cannot leak from a pot sealed at the top. And the lost weight is escaping water, not stored starch converted into vapour.

30. (D) The shaded leaf loses little water (low transpiration) but still makes starch in light (photosynthesis works) — so the two can vary independently and are distinct processes. The 'required/always together' option is contradicted by the very observation: low transpiration with successful food-making. 'Photosynthesis causes transpiration, so low loss means no food' is disproven because the low-loss leaf DID make starch. They are not the same process under two names — transpiration is water vapour leaving the stomata, photosynthesis is food being made; the data show one can be low while the other proceeds.

31. (A) Transpiration is the loss of water vapour through stomata, which then condenses as droplets in the bag. A living leaf's stomata work, so it transpires and droplets form; a dead leaf's cells no longer regulate or drive this loss, so far fewer droplets appear. The rain idea is excluded — the bag is sealed and identical for both, so any difference is the leaf itself. Xylem carries water inside the plant but does not 'squirt' water out of the surface; loss is as vapour through stomata. The dead-leaf option is backwards: a dead leaf does not MAKE water; it loses less because its transpiration has effectively stopped.

32. (B) Xylem carries water and dissolved minerals UPWARD from the roots, so claim 1 is correct. Phloem carries the FOOD made in the leaves outward and largely downward — it does NOT carry water upward, so claim 2 is wrong. Hence only claim 1 is correct. 'Both correct' accepts the false claim 2. 'Only claim 2' wrongly says xylem carries the food. 'Neither / both carry food' ignores that xylem's job is water transport. Exactly one claim — claim 1 — stands.

33. (D) The correct word equation is carbon dioxide + water → (sunlight, chlorophyll) → glucose + OXYGEN. The pupil wrote 'carbon dioxide' on the right where oxygen belongs — oxygen is the gas released. So the one fix is changing the right-hand carbon dioxide to oxygen. Changing left-hand water to oxygen breaks a correct raw material. Removing chlorophyll deletes a needed condition. Changing glucose to starch is wrong — glucose is the immediate product; starch is the later storage form. Only the right-hand-gas error needs correcting.

34. (A) Green plants are the great regulators of atmospheric gases. In photosynthesis they take in carbon dioxide and release oxygen — exactly reversing what animals' breathing and the burning of fuels do — so the proportions of the two gases stay roughly steady over the years. Respiration in plants does the OPPOSITE of balancing: like animals, respiring plants take in oxygen and give out carbon dioxide, so it cannot be what removes the excess CO₂. Transpiration is the loss of water vapour through stomata; it does not 'wash carbon dioxide out of the air.' And soil does not soak up the air's carbon dioxide through roots — roots take up water and minerals, not atmospheric CO₂.

35. (C) Photosynthesis needs light as its energy source. Plant A under the lamp has light and makes starch (turns blue-black); plant B in the dark box has CO₂, water, warmth and chlorophyll but NO light, so it makes none. Since light is the only factor that differed, the experiment isolates LIGHT. 'Both blue-black' ignores that B had no light. 'Only B, darkness concentrates starch' is backwards — no light means no starch. 'Neither' is wrong: one lit day is ample for plant A to make detectable starch.

36. (C) Water, warmth and carbon dioxide were held identical while only light differed, and growth climbed steadily with light (40 g in full sun down to 0 g near-dark). So more light drives more photosynthesis and hence more food and growth — light is the limiting factor here. 'No clear effect' ignores the obvious trend. 'Darkness grows plants fastest' misreads the 0 g near-dark plot. Carbon dioxide cannot be the cause because it was kept the same across all four plots — only light varied.

37. (D) The day's photosynthesis needs carbon dioxide, water, light and chlorophyll — not soil nitrogen — so the spinach was photosynthesising even while pale. The paleness and stunting, cured by nitrogen-rich manure, show nitrogen is needed to build proteins and the green pigment for healthy growth — but it is not a direct raw material of that day's food-making. So (D) is right. Nitrogen is not the raw material turned into starch, nor the carbon source (carbon comes from CO₂), and it does not replace sunlight.

38. (A) In the legume–Rhizobium partnership the BACTERIA fix air nitrogen into a usable form for the plant; in return the plant — a food-maker — gives the bacteria sugary food and a protected home in the root nodules. So what the bacteria receive is food and shelter. 'Usable nitrogen from the plant' reverses the deal — nitrogen flows from bacteria TO plant. The plant does not give chlorophyll — bacteria do not photosynthesise. And it does not hand them carbon dioxide to build glucose; the bacteria get ready-made sugars from the plant.

39. (C) Repeatedly cropping maize drains soil nitrogen. The natural, fertiliser-free remedy is to grow a leguminous crop (such as beans) for a season: the Rhizobium bacteria in its root nodules fix atmospheric nitrogen, and ploughing the crop back returns that nitrogen to the soil. Watering more heavily adds no nitrogen. Planting maize closer together uses the soil harder, draining nitrogen faster. Leaving the field bare for a year does not restore nitrogen by itself — nothing fixes the air nitrogen without the legume's bacteria.

40. (B) In a lichen the ALGA has chlorophyll and photosynthesises, making the food the pair lives on. So the food-maker you would point to is the alga. The fungus does not photosynthesise — it supplies water, minerals and shelter and takes food from the alga, so the options crediting the fungus are wrong. 'Neither, lives only on minerals' is wrong too — a lichen does need a food-maker, and that is the alga.

41. (D) Iodine turns blue-black only where STARCH is present. Boiled rice and raw potato are starch-rich, so they react strongly; an apple (mostly sugar) and boiled egg yolk (mostly fat and protein) have little or no starch, so they barely change. The reaction is about starch, not about being 'living/uncooked' — boiled rice is cooked yet reacts. The blue-black shows starch, not chlorophyll. And it has nothing to do with sweet taste — apple is sweet yet gives almost no colour.

42. (A) A broad, flat blade presents the largest possible surface to sunlight, and packing the chlorophyll-rich cells just beneath the lit upper surface puts the pigment where the light is strongest — both features serve efficient photosynthesis (light capture). The thin, flat shape is not built to store large amounts of water — water-storing organs are thick and fleshy. The broad shape catches wind rather than resisting it, so 'stays heavy so the wind cannot move it' is wrong. And leaf shape is not an anti-insect toughness adaptation.

43. (C) The pores are stomata. Their chief role in nutrition is to let carbon dioxide — a raw material of photosynthesis — enter the leaf, while oxygen and water vapour pass out through them. Veins are not these surface pores; the food made in the leaf travels in the phloem inside the veins, not out through the stomata. Root hairs absorb water from the soil, not gases at the leaf surface. And stomata do not 'absorb mineral nutrients from the air' — minerals come from the soil through the roots; the stomata's job is gas exchange. Both 'stomata' options name the structure correctly, but only one states the correct role.

44. (B) Food is made only in the green, chlorophyll-containing parts — mainly the leaves — never in a carrot's root or a potato's stem-tuber. The food made in the leaves is carried through the phloem to these storage parts, where the surplus is laid down as insoluble starch. So a storage organ does not make its own food (the green leaves did); the roots do not absorb ready-made food from the soil (that is the food-from-soil misconception); and water carried up in the xylem is a raw material, not something that 'turns into' stored food by itself.

45. (B) A brown (no-colour) iodine result simply means no starch is present; it says NOTHING about whether the tissue ever contained chlorophyll — so (B) is the FALSE statement. A blue-black colour does indicate starch is present — true. Destarching the plant beforehand does let any new starch be credited to the experiment — true. Boiling the leaf in alcohol does remove chlorophyll so the colour change is easy to read — true. Only (B) is false.

46. (D) Alcohol is highly flammable, so heating it directly over a flame risks catching fire. Standing the alcohol tube in a beaker of hot water heats it safely and evenly, which is why the test is done this way. Water is not 'added to the alcohol to dissolve chlorophyll' — the alcohol itself removes the chlorophyll. Heating does not turn alcohol into starch. And it is the heat, not any 'light from the flame', that the step needs — the point is simply to avoid a naked flame on flammable alcohol.

47. (C) The essential difference is the MODE of nutrition. The banyan is an autotroph: it builds its own food from carbon dioxide and water using sunlight. The goat is a heterotroph: it cannot make food and must consume ready-made food. The banyan does not eat small soil creatures, nor does it make food 'inside its roots' — food is made in green tissue from CO₂ and water. 'Both take ready-made food from their surroundings' is false for the banyan, which manufactures its own.

48. (B) The bread mould has no chlorophyll, so it cannot make its own food by photosynthesis — ruling out the kitchen-light and the warmth-and-moisture options. Instead it feeds as a saprotroph: it secretes digestive juices onto the bread, which break the food down outside the fungus, and then it absorbs the dissolved food. It does not trap and digest insects — that is an insectivorous plant's method, not a bread mould's. So option (B) correctly describes saprotrophic nutrition by external digestion and absorption.

49. (D) *Cuscuta* thrives on a LIVING hibiscus but shrivels and dies on a lifeless plastic stake — exactly the behaviour of a parasite, which can feed only by drawing food from a living host. A saprotroph feeds on dead, decaying matter, which a plastic stake is not, and it would not depend on a living shrub. A green autotroph would make its own food and not die for lack of a host; *Cuscuta* is genuinely chlorophyll-poor, not 'briefly' discoloured. A symbiont benefits its partner; *Cuscuta* harms its host. So it is a parasite.

50. (B) Both facts must fit one explanation. The sundew's green leaves photosynthesise to make its energy-rich food; it traps and digests insects mainly to get the scarce NITROGEN its boggy soil cannot supply. So it is a green food-maker that supplements nitrogen by catching insects. The green leaves are not merely ornamental — they genuinely photosynthesise. It is not a parasite — it does not draw food from a living host. And it is not a saprotroph — it does not live on dead, decaying matter; its green leaves are functional.

51. (D) With photosynthesis clearly outpacing respiration in steady bright light, the plant releases more oxygen than it uses and takes in more carbon dioxide than it releases. So by the end oxygen has RISEN and carbon dioxide has FALLEN. 'O₂ fallen, CO₂ risen' describes net respiration (darkness), the opposite case. 'Both risen' is impossible when CO₂ is being consumed. Nitrogen is not taken in or released by either process, so it does not fall sharply.

52. (C) A green plant begins a food chain because it captures sunlight to create brand-new food, introducing fresh energy into the chain — it is a producer. A mushroom only breaks down food (dead matter) that already exists; it adds no new energy, so it cannot be the starting point. The green plant is not 'the only thing that respire' — all the organisms respire. The mushroom does not eat the plant to begin the chain. And size has nothing to do with which organism is the producer.

53. (D) A tiger eats only meat, but the deer it eats fed on plants — so the energy in the tiger's food was first captured from sunlight by green plants and passed up the chain. Plants also release the oxygen the tiger needs to release that energy by respiration. That is the deep, universal dependence. Occasional grass-nibbling when ill is trivial and not the reason. Plants do not pass soil minerals 'directly' to the tiger. And plants release most of their oxygen in daylight, not specially at night.

54. (C) Light is the energy source for the algae's photosynthesis. Persistent shade starves them of light, so they make little food and decline (producers fail first). The snails, which depend on the algae for food, then run short and decline too — the impact passes up the chain from the base. 'Snails thrive on softer algae' invents a benefit; less light means less algae, not better food. Algae cannot 'feed on silt' — they are autotrophs that need light, not particles, to make food. 'Algae don't need light' is flatly false; without light photosynthesis stops.

55. (A) Autotrophs make their own food, but not from nothing — they still need raw materials and energy from outside: carbon dioxide, water and especially light. Locked in a dark cupboard a green plant cannot photosynthesise, so it cannot feed itself and will starve once its reserves are gone. 'Need nothing once grown' is exactly the misconception being corrected. The plant cannot survive by eating the cupboard's wood, nor by taking ready-made food from the air — those are not how a green plant feeds.

56. (B) Carbon dioxide is a raw material of photosynthesis. The KOH removes it from one leaf's air, so that leaf cannot make starch and stays brown; only the normal-air leaf makes starch and turns blue-black. The student's error was treating CO₂ as something whose removal HELPS, when in fact CO₂ is essential and its removal stops food-making. 'Both turn blue-black, light alone is enough' ignores that CO₂ is needed too. 'Neither, destarching destroyed the ability' misunderstands destarching, which only empties stored starch and does not disable photosynthesis. 'KOH feeds the leaf' inverts KOH's role — it ABSORBS CO₂, it does not provide it.

57. (D) The leaf releases net oxygen by day (photosynthesis exceeds respiration) and takes in net oxygen by night (only respiration). At the two zero-crossing instants around dawn and dusk, the rate of oxygen made by photosynthesis exactly equals the rate of oxygen used by respiration — the compensation point — so the NET exchange is zero while both processes still run. The leaf has not stopped both processes — respiration never stops. It is

not 'only respiring' or 'photosynthesis completely stopped' at those instants — those describe the night, which would give a net oxygen UPTAKE, not zero.

58. (A) Saprotrophs such as fungi and bacteria feed on dead leaves, wood and animal remains by breaking them down — and in doing so they release the nutrients locked inside that dead matter back into the soil, where green plants can absorb and reuse them. That recycling is why decomposers matter even though they make no food themselves. They do not photosynthesise or pass on sun-made food (they have no chlorophyll), and they are not the source of plants' oxygen — green plants make their own oxygen in photosynthesis. 'No real importance' ignores their essential role in returning nutrients to the soil.

59. (A) A plant's body is largely made of food (glucose/starch and related substances) whose carbon comes from carbon dioxide fixed in the leaves during photosynthesis — so the great bulk of new solid mass is built from a gas in the air, not solid soil. Roots take up water and dissolved minerals, not whole solid soil that hardens into wood, so the first wrong option misdescribes how roots work. Oxygen is a PRODUCT released by photosynthesis, not a raw material the tree turns into solid wood. The girl's whole-soil idea is the food-from-soil misconception the chapter overturns: the trunk's mass traces mostly to carbon dioxide, with water supplying hydrogen.

60. (B) Xylem carries water and dissolved minerals UPWARD from the roots; phloem carries the food (sugars) made in the leaves AWAY from them, largely downward. So the correct match is water/minerals = xylem/up and food = phloem/away-and-down. The first wrong option swaps the two tissues. 'Both in the xylem' is false — the plant has two distinct transport tissues. The last option reverses both directions (water down, food up). Only this row keeps each tissue with its correct cargo and direction.